Chapter 3, Problem 83P

Problem

The following program is the Schematics Netlist of a particular circuit. Draw the circuit and determine the voltage at node 2.

```
R_R1   1  2  20
R_R2   2  0  50
R_R3   2  3  70
R_R4   3  0  30
V_VS   1  0  20V
I_IS   2  0  DC    2A
```

Step-by-step solution

Step 1 of 4

From the SPICE code, resistor R_1 is between nodes 1 and 2. Resistor R_2 is between node 2 and ground. Resistor R_3 is between node 2 and node 3. Resistor R_4 is between node 3 and ground. Voltage source V_S is between node 1 and ground. Current source I_S is between node 2 and ground. Draw the circuit diagram.

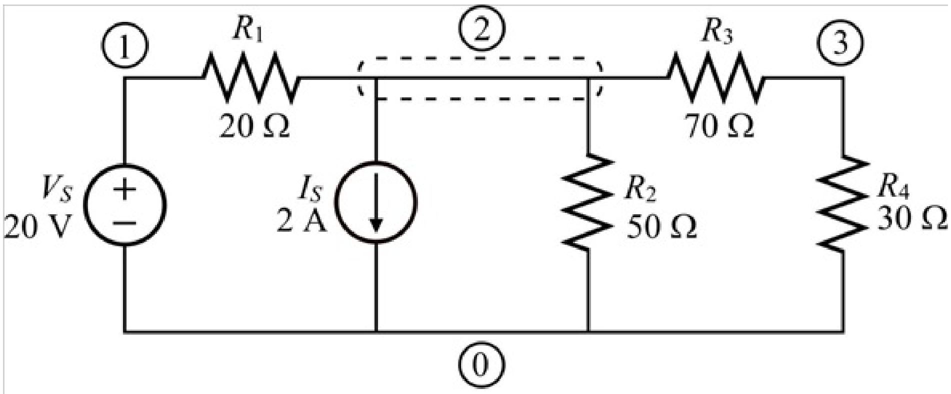

Figure 1

Step 2 of 4

From the circuit shown in Figure 1, the voltage at node 1 $V_1 = 20\text{ V}$.

Apply Kirchhoff's current law at node 2.

$$\frac{V_2 - V_1}{20\ \Omega} + 2\text{ A} + \frac{V_2}{50\ \Omega} + \frac{V_2 - V_3}{70\ \Omega} = 0$$

$$\frac{V_2 - 20\text{ V}}{20\ \Omega} + 2\text{ A} + \frac{V_2}{50\ \Omega} + \frac{V_2 - V_3}{70\ \Omega} = 0 \qquad [\text{since } V_1 = 20\text{ V}]$$

$$\left(\frac{1}{20\ \Omega} + \frac{1}{50\ \Omega} + \frac{1}{70\ \Omega}\right)V_2 - \frac{V_3}{70\ \Omega} = \frac{20\text{ V}}{20\ \Omega} - 2\text{ A}$$

$$\left(\frac{1}{20\ \Omega} + \frac{1}{50\ \Omega} + \frac{1}{70\ \Omega}\right)V_2 - \frac{V_3}{70\ \Omega} = 1\text{ A} - 2\text{ A}$$

Multiply with 70 on both sides of the equation.

$$(70)\left(\frac{1}{20\ \Omega} + \frac{1}{50\ \Omega} + \frac{1}{70\ \Omega}\right)V_2 - (70)\left(\frac{V_3}{70\ \Omega}\right) = (1\text{ A} - 2\text{ A})(70)$$

$$(70)(0.0843)V_2 - V_3 = (-1\text{ A})(70)$$

$$5.9V_2 - V_3 = -70\text{ V} \quad \text{...... (1)}$$

Step 3 of 4

Apply Kirchhoff's current law at node 3.

$$\frac{V_3 - V_2}{70\ \Omega} + \frac{V_3}{30\ \Omega} = 0$$

$$-\frac{V_2}{70\ \Omega} + \left(\frac{1}{70\ \Omega} + \frac{1}{30\ \Omega}\right)V_3 = 0$$

$$-\frac{3V_2}{210\ \Omega} + \left(\frac{3+7}{210\ \Omega}\right)V_3 = 0$$

$$-3V_2 + 10V_3 = 0$$

$$V_3 = 0.3V_2 \quad \text{...... (2)}$$

Step 4 of 4

Calculate the voltage at node 2.

Recall equation (1).

$$5.9V_2 - V_3 = -70\text{ V}$$

Substitute $0.3V_2$ for V_3.

$$5.9V_2 - 0.3V_2 = -70\text{ V}$$

$$5.6V_2 = -70\text{ V}$$

$$V_2 = \frac{-70\text{ V}}{5.6} = -12.5\text{ V}$$

Therefore, the voltage, V_2 at node 2 is $\boxed{-12.5\text{ V}}$.